



The Carnation Revolution and economic crisis: A synthetic control analysis (1974-2000)[☆]

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ABSTRACT

On 25 April 1974, the “Carnation Revolution” toppled Western Europe's oldest dictatorship, Portugal's *Estado Novo*. Political instability ensued, leading to a wage explosion, expropriations, and the collapse of an empire that brought back half a million repatriates. As a result, the Portuguese economy slowed from the high growth it had experienced in the 1950s and '60s. However, disentangling the effects of the revolution from those of the broader international crisis of the 1970s is challenging. To address this, we apply the synthetic control method and find that the Carnation Revolution, along with the ensuing instability, caused a negative structural break that persistently reduced GDP per capita. By 1976, GDP per capita was almost 15% lower than the counterfactual estimate. This coincided with a decline in the investment-to-employment ratio and higher current account deficits than those observed in comparable countries.

1. Introduction

On 25 April 1974, a military coup that went down in History under the name of “Carnation Revolution” toppled what was then Western Europe's oldest authoritarian regime, Portugal's *Estado Novo*. The country entered a complex political situation typical of the Cold War era, with political agents favoring liberal democracy and a market economy opposed by others advocating a socialist regime, the latter implying authoritarianism and a socialized economy. It took more than one year to define the political direction of the country in favor of democracy. However, many of the changes introduced during the confrontational period persisted and had a lasting impact on economic growth.

The legacy of the period in which the socialist project gained the upper hand included capital flight and a wage explosion, at a time when the country had to rapidly absorb a large influx of repatriates coming from former colonies. Assessing the impact of these factors is challenging, as they coincided in time with the 1970s international crisis resulting from the end of the Bretton Woods system and the oil price shock triggered by the Arab countries of the Organization of Petroleum Exporting Countries (OPEC) embargo in October 1973

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(Crafts and Toniolo, 1996; Eichengreen, 2007). All Western economies experienced a period of slowdown and external imbalance. However, we argue that the crash was more severe in Portugal due to the revolution and the instability that followed. The purpose of our paper is to disentangle the portion of the economic downturn and imbalance attributable to the international crisis from that caused by the Carnation Revolution, and to assess the long-term consequences of the latter. To do this, we use the synthetic control method, comparing the evolution of the Portuguese economy with a synthetic economy composed of countries that best replicate the Portuguese economy's trajectory before 1974 (Abadie, 2021). We apply this method to the period 1955–2000, using 1955–1973 as the pre-treatment period and 1974–2000 as the treatment period, to capture both short- and long-term effects.

Our findings can be summarized as follows. First, the combination of an economic crash with a severe balance-of-payments crisis created a fundamental dilemma for the post-revolutionary Portuguese economy: efforts to revive growth often led to external imbalances, while attempts to correct those imbalances suppressed growth. Second, capital flight and a labor shock, both directly resulting from the revolution, permanently altered the factor mix of the economy, making it less capital-intensive and affecting its long-term growth path. As a result, the Portuguese economy underwent a structural shift: until 1973, it was one of the quickest growing economies in the world; from then on, its performance was comparatively mediocre.

We propose that the Carnation Revolution and its economic consequences should be understood as a case-study within the framework of the theory of revolutionary outcomes (Goldstone et al., 2022; Goldstone, 2023), comprising the analysis of the inevitable instability following the demise of authoritarian regimes (Kuran, 1989). A key point in this literature is that revolutionary outcomes often diverge from the intentions of revolutionary agents. As noted by Goldstone (2023, p. 1): “the overthrow of the existing government is often unexpectedly easy, as fragile regimes can collapse in a matter of days or weeks. The more difficult task is for the new revolutionary regime (...) to achieve its expressed goals, which often include greater economic prosperity, a more egalitarian distribution of rewards, ending corruption, and a less oppressive state”. While the last of these objectives was achieved by the Carnation Revolution, the same cannot be clearly said of the first: the initial steps of the revolution contributed to a severe crisis and diminished the growth potential of the new democracy. In Kuran's (1989) perspective, this results from the fact that during authoritarianism people engage in what he calls “preference falsification”, i.e. they do not show their true political preferences. When revolution erupts, preferences that were hidden (or falsified) come to the light and become explicit, revealing the myriad reasons why people were unsatisfied with the authoritarian regime. Once these now made explicit political preferences clash with each other, instability or uncertainty is the natural result.

The question of revolutionary outcomes has received much more attention from recent literature than that of their onset. For Acemoglu and Robinson (2006), Goldstone et al. (2022), and Goldstone (2023) the issue of why revolutions lead to different outcomes is of crucial relevance, as some result in authoritarian regimes replacing the previously existing authoritarian ones, while others lead to the establishment of constitutional democracies. In doing so, this literature tries to overcome the difficulties of traditional rational choice theory, for which revolutions are paradoxical, to the point of being almost impossible to explain, as individual rational decision-making, according to its assumptions, should never result in collective revolutionary action (Olson, 1965; Tullock, 1971; Courtois et al., 2024).

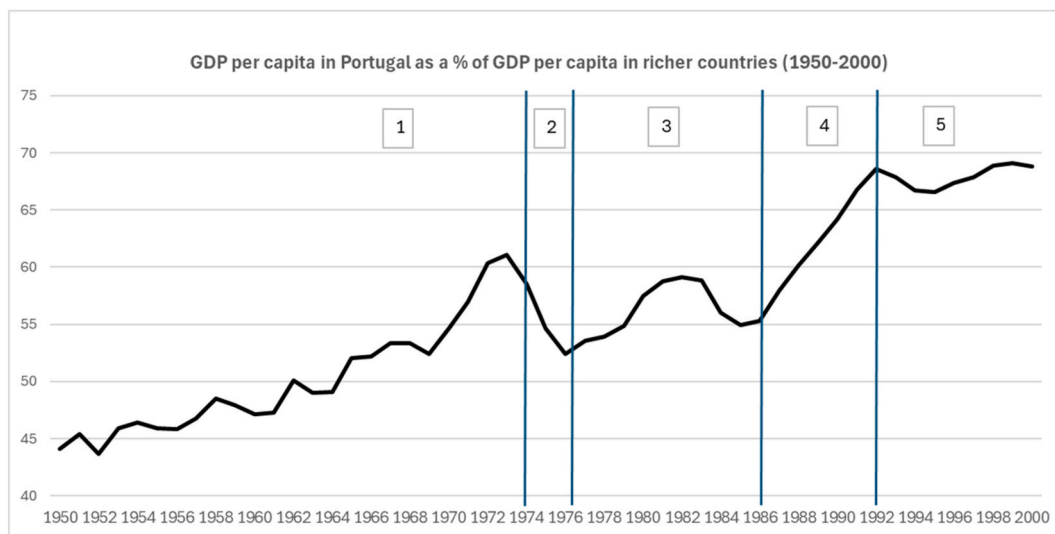


Fig. 1. GDP per capita in Portugal as a % of GDP per capita in richer countries (1952–2000)

Notes: countries represented in the sample: Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Italy, New Zealand, Norway, Sweden, Switzerland, the UK, and the US. Time periods: (1) 1950–1974: strong ‘catch-up’ growth; (2) 1974–76: precipitous decline in the wake of the disruptions brought by the Carnation Revolution; (3) 1976–1986: stagnation as a result of lingering post-revolutionary ‘hangover’ effects, and (4) 1987–1992: resumption of strong ‘catch-up’ growth. I think these might be delineated even more strongly on the graph by vertical partitions; (5) renewed stagnation in convergence.

Sources: Portugal – INE and BdP (2021) and Amaral (2009); for the other countries: The Conference Board. Unit: 2011 PPP US dollars

Goldstone (2023) argues that much of the revolution's outcomes depend on whether it is initiated by radicals or moderates. We propose that the Carnation Revolution was distinct, being initiated by both. What followed was indeed a choice between radicalization and moderation, at the end of which moderation ultimately prevailed. But we show that its victory was accompanied by the incorporation of a radical legacy that had a significant and lasting impact on the future of the democratic regime and on its economic prospects. As a case-study in revolutionary outcomes, the Carnation Revolution illustrates that such outcomes need not be strictly binary; they can combine elements of both radicalization and moderation. This fits well into Kuran's (1989) perspective of a necessarily intricate situation after revolutions, precisely leading to a possible final outcome of compromise between radicalization and moderation.

This paper is organized as follows. Section 2 reviews the main features of the Portuguese economy from the post-war period to the revolution and describes the national and international context of the 1970s crisis. Section 3 outlines the synthetic control method and the data used. Section 4 presents the results and Section 5 concludes.

2. The Portuguese pre-revolutionary economy, the 1970s international crisis, and the revolutionary crisis

2.1. The Portuguese pre-revolutionary economy

The period between the end of World War II and 1973 was the most successful in terms of growth in Portugal's history. The country experienced sustained convergence, especially during the 1960s (Fig. 1). In 1950, Portuguese GDP per capita was 45% of the average of the richer economies; by 1973, it had risen to 62%, reducing the gap by nearly 20 percentage points. The process occurred despite quick growth in most advanced economies in the context of the "golden age of economic growth" (Crafts and Toniolo, 1996).

High growth in Portugal resulted from a favorable combination of conditions for accumulating both physical and human capital, as well as significant gains in economic efficiency. These conditions for physical capital accumulation stemmed from several factors that provided the country with abundant investment funds. First, Portugal's neutrality during World War II enabled the accumulation of substantial gold and foreign exchange reserves, which were recycled into the financial system to lower credit costs (Amaral, 2005). Abundance of funds persisted, first, in the 1950s, thanks to the Korean War (1950-1953), during which many Portuguese exports were highly demanded, and then in the 1960s, thanks to the influx of an unprecedented number of emigrant remittances, corresponding to an equally unprecedented wave of emigration (Amaral, 2005).

Human capital accumulation advanced driven by a combination of increased returns to education (due to growing physical capital) and policies that expanded primary education in a country where illiteracy was still very high. This led to a steady increase in primary school enrolment and a noticeable decline in the illiteracy rate (Palma and Reis, 2021; Palma, 2023). Although the adult illiteracy rate remained at 25% in 1974, universal enrolment of children under the age of 12 had already been achieved by 1960 (Amaral, 2005; Gomes and Pinto Machado, 2020).

Efficiency gains were primarily the result of growing exposure to international competition, driven by European economic integration and trade liberalization. The *Estado Novo* regime actively participated in major postwar international cooperation initiatives. The most significant was Portugal's membership in the European Free Trade Association (EFTA) in 1960 as a founding country, alongside the United Kingdom, Switzerland, Austria, Denmark, Norway, and Sweden. This step was later complemented by a free trade agreement with the European Economic Community (EEC) in 1972 (Leitão, 2007; Amaral, 2005). These stronger ties with Western European economies forced Portugal to reduce protectionist policies and compete with more efficient producers, which in turn boosted total factor productivity (Lains, 2003; Amaral, 2005).

These economic achievements took place under the authoritarian regime of the *Estado Novo*, which started to feel international pressure to change during the 1960s and 1970s. Some authors have suggested that late authoritarian regimes may initiate liberalizing reforms to prevent political destabilization (Prados de la Escosura et al., 2011; Nowrasteh et al., 2020). In Portugal, Marcello Caetano replaced António de Oliveira Salazar in 1968 as the strongman of the regime and introduced limited institutional reforms, often referred to as the "Marcellist Spring." However, these reforms were gradual, administratively constrained, and did not amount to a comprehensive stabilization or liberalization program comparable to Spain's 1959 Stabilization Plan. Moreover, Portugal's main trade liberalization had already occurred through EFTA membership in 1960 and subsequent integration agreements. Empirically, no structural break in growth or capital accumulation is visible prior to 1974, suggesting that the underlying growth regime remained intact until the revolution.

By the time of the revolution, the most prominent political proposals for change reflected the two dominant ideological models of the time: a transition to democracy, similar to those existing in Western Europe at the time, or a shift to a communist/socialist regime akin to those existing in Eastern Europe (Reis, 1994; Ramos et al. 2009; Amaral, 2005). Within the framework of the theory of revolutionary outcomes, the former can be seen as representing the moderate side of the revolution, and the latter the radical side. Portugal's situation was further complicated by the fact that the regime was engaged in a colonial war on three fronts: Portuguese Guinea, Angola, and Mozambique, where independentist movements had started guerrilla activities to obtain independence since 1961.

2.2. A revolution within an international crisis

The economic path described above was interrupted by the 1974 Carnation Revolution, a military coup that toppled the *Estado Novo*. The coup was initiated by a group of junior military officers, organized into an association they named the Armed Forces Movement (*Movimento das Forças Armadas*, MFA), determined to end the Colonial War. However, they linked the end of the war to the

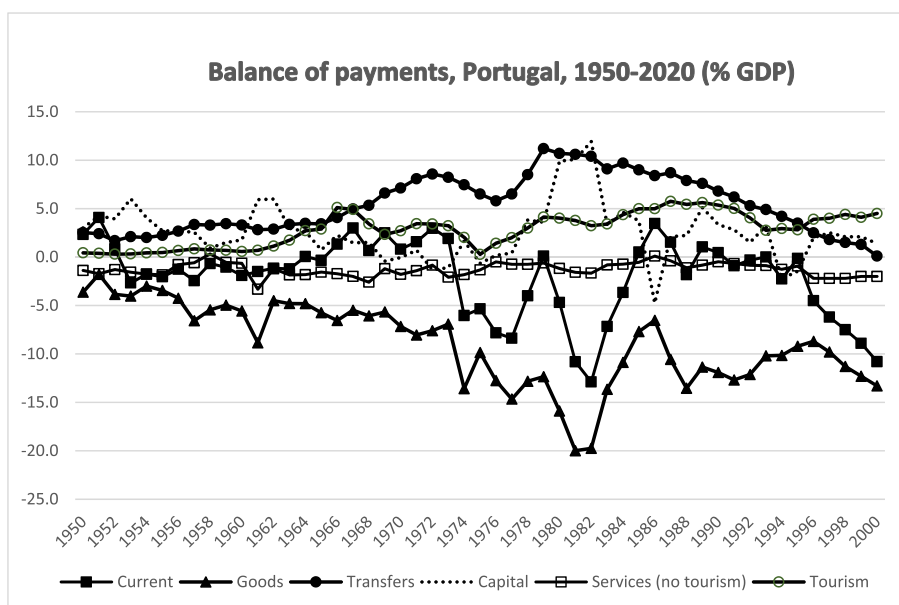
overthrow of the regime itself. On 25 April 1974, with the support of some senior officers, the MFA orchestrated a coup that brought down the regime and initiated a transitional process to establish a new political order (Reis, 1994; Ramos et al. 2009; Amaral, 2005).

There was no consensus among the revolutionaries regarding the nature of the new regime. They split over the issue, together with the political actors and the society of the time. The division followed a typical Cold War pattern, with some favoring a democracy and a market economy, while others favored a socialist revolution (Reis, 1994; Ramos et al. 2009; Amaral, 2005). Initially, the factions aligned with more radical socialist interpretations gained the upper hand (Reis, 1994; Ramos et al. 2009; Amaral, 2005). Labor unrest played a major role in this radicalization, with the occurrence of various strikes and many episodes of occupation of firms by workers. With the revolutionary authorities abstaining from repressing these actions, wages increased 7% and 14% in real terms in 1974 and 1975, respectively (Mateus, 2006). This wage surge was accompanied by a permanent increase in labor costs, driven by new legislation introduced by successive governments. Measures included the establishment of a national minimum wage, the legalization of strikes, the creation of a general system of unemployment benefits, the liberalization of union activity, and the enactment of a new Labor Code, which made individual dismissals nearly impossible and collective layoffs very difficult (Amaral, 2005).

At the same time, the labor market experienced a massive shock resulting from the decolonization process. Negotiations to grant independence to the Portuguese colonies began soon after the 1974 coup and panic spread among the Portuguese colonists in Africa, who began an exodus toward the Portuguese mainland. In one of the largest population movements ever in Portuguese history, an estimated 500,000 to 700,000 people (approximately 8% of the mainland population) entered the country between late 1974 and early 1976 (Pires et al., 1984; Carrington and de Lima, 1996; Mäkelä, 2017). These *retornados* were, on average, more educated and more entrepreneurial than the native population, bringing disproportionately high levels of human capital into the Portuguese labor market (Bohnet et al., 2025).

Thus, at a time when both the international crisis and the labor supply shock should have exerted downward pressure on wages, the opposite occurred. As noted, real wages rose 7% and 14% in 1974 and 1975, but according to the calculations provided by Krugman and Macedo (1981), they should have fallen by 10% under the prevailing circumstances. Consequently, firms already burdened by rising energy costs and the increasing price of imported goods due to the 1973 oil shock, found themselves constrained by unprecedented wage pressures. Capital flight began then (Amaral, 2005) and intensified after the revolutionary authorities nationalized the assets of large business groups and private firms operating in sectors deemed "strategic", in one of the largest nationalization programs ever in Western Europe (Amaral, 2005; Valério, 2004). As a result, the Portuguese state-owned entrepreneurial sector became one of the largest in the Western world, with the Government controlling totally or partially a wide range of sectors: currency issuance, banking, insurance, basic metals, shipbuilding and repair, cement, paper and paper pulp, chemicals and petrochemicals (Baklanof, 1992).

These developments led to a severe economic crash, disrupting the path of rapid growth and convergence that had defined the previous three decades (Fig. 1). The downturn was further exacerbated by a sharp deterioration in the balance of payments, as shown in Fig. 2. The current account moved from a small surplus in 1973 to a deficit of 5–6% of GDP in 1974 and 1975. This combination of economic collapse and massive external imbalance became the country's central economic challenge (Krugman and Macedo, 1981). As Schmitt (1981, p. 1) observed, "the problem of managing economic growth with a balance of payments constraint was new to



Source: INE and BdP (2021)

Fig. 2. Balance of payments, Portugal, 1950-2020 (% GDP).

Source: INE and BdP (2021)

Portugal.”

2.3. Persistent external imbalance

The political turmoil following the Carnation Revolution eventually subsided by the end of 1975. The first elections after the revolution were won by political forces favoring a Western-style democracy and market economy, defeating those advocating for a socialist regime. Although the socialist plans were abandoned, they left a significant legacy in the form of widespread nationalization of the main sectors of the economy and massive capital flight. Once the decision to align the Portuguese economy with the Western bloc of the Cold War prevailed, successive governments from 1976 onward faced the dual challenge of addressing the economic collapse of the previous two years while simultaneously implementing Welfare State social programs, which had been largely underdeveloped under the *Estado Novo*. These objectives often proved difficult to reconcile due to persistent external imbalances. Efforts to stimulate growth or expand social programs placed additional pressure on the balance of payments, creating a dilemma that was hard to resolve. The main instruments used to pursue both economic recovery and social development were, on the one hand, an expansion of public spending, which became seriously unbalanced, and on the other, a series of investment programs leveraged on the now vast arsenal of public companies (Lopes, 2005; Amaral, 2005). The result was a full decade of alternating periods of stop-go, with external imbalance resulting from growth periods, inevitably followed by austerity aimed at restoring balance.

It was in these circumstances that the Portuguese authorities negotiated three International Monetary Fund (IMF) interventions, in 1977, 1978, and 1983 (Zorrinho, 2018; Amaral et al., 2020). The results of the IMF programs were very effective in terms of external rebalance, visible in the return to balance of the current account in 1979, as Fig. 2 shows (details in Amaral et al., 2020), though they came at the cost of reduced economic growth. Nevertheless, by 1985, fundamental external balance had been achieved, with the conclusion of the third IMF program. This marked the end of Portugal's transition from revolution to democracy, a milestone further confirmed by the country's accession to the EEC in 1986 (Cunha, 2015).

The EEC project, however, was changing substantially then, as it was entering a crucial new phase, marked by the adoption of both the Single Market and Monetary Union programs. In February 1992, the European Council met in Maastricht to sign a treaty that not only created the European Union (replacing the EEC) but also laid the groundwork for the creation of a single currency, scheduled for completion by 1999 (James, 2012). Portugal, having decided to join the monetary union, implemented a monetary policy in line with this goal (Abreu, 2001). The nominal value of the Portuguese currency (the *escudo*) remained broadly stable relative to those of its major trading partners from 1992 onward, with only small depreciations of 6.5% in 1993 and 3.5% in 1995 to correct overvaluation. Portugal was thus made eligible to join the European Monetary Union (EMU), becoming one of its founding members in 1999. Participating in the EMU – but also the entry of China in the World Trade Organization – had major economic impacts on the Portuguese economy, marking strong enough breaks to be considered by themselves, separately from the Carnation Revolution (Gabriel and Pessoa, 2024; Cabral et al., 2021). For this reason, we stop this exercise at the time of its creation.

To sum up, we propose that the sequence of events between 1974 and 1985, as described above, produced an economic crash and a balance of payment crisis that was more severe in Portugal than in most Western countries, creating an economic policy dilemma: if governments chose to stimulate growth to lift the economy from the crash and to develop social policies popular among voters, they would further deteriorate external balance; if, instead, they tried to correct external balance, they risked jeopardizing recovery from the crash and the development of social policies. The difficulties of policymakers to square the circle brought three IMF interventions, in 1977, 1978, and 1983. It was only after these interventions that both objectives could be achieved, in 1985, eleven years after the revolution.

By that time, however, the growth potential of the economy had been seriously affected by the revolution and the instability that followed. This impact manifested in two key dimensions: a labor shock, resulting from the mass influx of repatriates combined with a wage explosion; and a negative capital shock, stemming from capital flight and subsequent capital scarcity following the widespread nationalizations of 1975.

The factors that had driven the rapid growth of the Portuguese economy from the 1950s to 1973 had largely disappeared. Funds for investment became scarce due to external imbalances and capital flight, despite continued high levels of emigrant remittances. Fiscal policy and inefficient investment programs absorbed much of the remaining capital, diverting it from more productive uses. Of the favorable conditions for growth that existed between 1950 and 1973, only human capital accumulation continued apace, reflecting the priority the new authorities placed on expanding the public education system (Amaral, 2005). But this was not enough to compensate for the other factors, since human capital growth was the lesser of the contributors to growth in the Golden Age, as most available growth accounting exercises seem to confirm (Lains, 2003; Amaral, 2005).

The simultaneous occurrence of the revolution and of the 1970s international crisis creates an observational problem: how much of Portugal's economic decline can be attributed to the revolution, and how much to the international context? To isolate the respective impacts, we use a statistical method that has been increasingly applied to measure the effect of certain events or policy interventions on economic outcomes, the synthetic control method (SCM). This method was introduced by Abadie and Gardeazabal (2003), and further developed by Abadie et al. (2010, 2003) and Abadie (2021). Athey and Imbens (2017, p. 9) consider the SCM to be “arguably the most important innovation in the policy evaluation literature in the last 15 years”, and it has been applied to assess the economic impact of events such as terrorism in the Basque country (Abadie and Gardeazabal, 2003), German reunification in 1990 (Abadie et al. 2010), the introduction of the European Monetary Unit (Hope, 2016), the 1953 London Debt Agreement (Galofré-Vilà et al., 2019), separatist governments in Quebec (Geloso and Grier, 2022), oil and gas sector nationalizations in South America in the early 2000s (Lucke and Rehfeldt, 2023; Thailand's 2006 military coup (Suwanprasert, 2023), Brexit in 2016 (Born et al., 2019; Venâncio and Pereira dos Santos, 2024) or, closer to the topic of this paper, the mass influx of white settlers returning to Portugal following the independence of

the former Portuguese colonies in 1975 (Mäkelä, 2017). We implement three synthetic control tests, applying them to GDP per capita, the current account balance, and investment per employed person, all between 1974 and 2000, as detailed in the following section.

Seen under the light of the theory of revolutionary outcomes (Acemoglu and Robinson, 2006; Goldstone et al., 2022; Goldstone, 2023) we propose that the Carnation Revolution should be understood as a variety not clearly identified in the theory. According to the latter, the outcome of any revolution depends on whether it is initiated by radicals or moderates. We propose that the Carnation Revolution was initiated by both, something that led to a confrontation over the issue of what type of regime would be finally installed: a radical (socialist) one or a moderate (liberal democratic) one. In the end, moderation prevailed, but its victory was accompanied by the incorporation of a radical legacy that had a lasting impact on the future of the democratic regime and on its economic prospects. As a case-study in revolutionary outcomes, the Carnation Revolution illustrates that such outcomes need not be strictly binary; they can combine elements of both radicalization and moderation, and that the former can affect the success of the (ultimately) moderate change. This fits well into Kuran's (1989) perspective, where results of that type are explainable by the fact that during authoritarianism people engage in what he calls "preference falsification", i.e. they do not show their true political preferences. When revolution erupts, preferences that were hidden (or falsified) come to the light and become explicit, revealing the myriad reasons why people were unsatisfied with the authoritarian regime. Once these now made explicit political preferences clash with one another, instability or uncertainty results, with a possible compromise between radicalization and moderation being achieved in the end.

This outcome is particularly interesting in the context of recent "populist" political experiences, particularly in Latin America (Grier and Maynard, 2016; Cachanosky and Padilla, 2019a, b; Absher et al., 2020; Cachanosky et al., 2025). What the Carnation Revolution shows is that radical economic changes of a populist type conducted by revolutionary agents leave an imprint on the country's institutional fabric, even if in the end moderation prevails. In this sense, a revolution like the Carnation Revolution is different from the 1959 Cuban one, where in the end radicalization prevailed over moderation and, hence, the economic impact was extremely negative (Jales et al., 2018; Bologna Pavlik and Vincent, 2025). But it could be closer to experiences such as those that have recently happened in Argentina, Venezuela or Ecuador (Cachanosky et al., 2025), where changes did not go as deep as in the case of Cuba, but still could have a long-term impact and affect durably the future of those countries' economies, with possible political consequences.

The Carnation Revolution also shows the importance of the insertion of the country undergoing revolutionary change within an international institutional environment harnessing the said country to a set of moderating institutional solutions. This is clear in the exchange the Portuguese authorities had from a certain point onwards with the IMF, whereby the country adopted a series of measures limiting the furthering of the radical revolutionary proposals. This can be seen as having some similarity with more demanding solutions, such as the adoption of the monetary institutions of other countries to deal with the institutional fragilities of countries undergoing radical change, as discussed in Cachanosky et al. (2022). A solution of that type may not entirely undo the effects of the revolutionary changes but can provide at least a means to limit their impact.

The primary objective of our paper is to assess the impact of the revolution on Portuguese economic growth, particularly what caused the shift from the rapid, "Asian Tigers"-style growth up to 1973 to the significantly slower pace thereafter. While global growth declined after 1973, Portugal's downturn was notably more pronounced. We attribute this to two main factors. First, a long-term decline in capital intensity, a specific effect of the revolution due to a combination of capital flight with a massive labor shock. And second, the external constraint, as expressed in the current account, a fundamental limitation for a small economy like Portugal. While we could have tested individual components contributing to the external imbalance (such as public spending deficits tied to the Welfare State, inflation differentials with trade partners, or the role of public investment), the current account summarizes all the above tensions, as it reflects not only those internal pressures but also other factors such as declining global demand for Portuguese exports, increased import dependency, the contraction of remittances, and reductions in foreign currency inflows from tourism. The current account thus captures both internal and external tensions and represents a constraint the economy struggled to overcome: attempts to stimulate growth worsened external imbalances, while efforts to correct them inevitably slowed down growth.

We do not include human capital in our set of synthetic control tests because human capital continued to grow steadily both before and after the revolution. The problem was that this was not sufficient to sustain previous levels of economic performance. Analyzing a variable that performed well to explain one that did not would not be logical. Clearly, human capital was not the cause of the post-1973 slowdown in Portugal's growth path.

3. The synthetic control method and the data used

The synthetic control method proposes a way of comparing a "treated unit" with a "donor pool". This process follows two main steps. In the first, the treated unit (in our case, the Portuguese economy) is compared with the donor pool (e.g., a group of relevant, comparable economies) over a pre-treatment period. The goal is to identify the weighted combination of donor units that best replicates the behavior of the treated unit during that period. In the second step, after the treatment (in our case, the Carnation Revolution and the instability that followed) has taken place, we assess how the behavior of the treated unit differs from the behavior of the synthetic series. The post-treatment evolution of the donor pool serves as a counterfactual, that is, an estimate of how the treated unit would have evolved in the absence of the event.

Formally, consider a sample of $J + 1$ units (national economies, in the case of this study) are considered. Unit $j = 1$ is the "treated unit" Portugal and units $j = 2, \dots, J + 1$ are the "donor pool" potential units to compare. In order to avoid interpolation bias, the units that are part of the donor pool should be chosen from among units having similar characteristics in terms of structural processes: in the case of this paper, the donor sample comprises OECD countries. Relying on these countries has the additional benefit that OECD countries tend to have high-quality, consistent, and comprehensive economic data. All units are observed in the same time periods, $i = 1, \dots, T$. There is a pre-treatment period, defined as T_0 , and a post-treatment period, defined as T_1 , the overall time period being $T =$

$T_0 + T_1$. The treatment does not affect unit 1 in T_0 , affecting it only during T_1 .

The synthetic control should be understood as a weighted average of the units in the donor pool and is defined by a $(J \times 1)$ vector of weights $W = (w_2, \dots, w_{J+1})$, with each weight restricted to being positive and having a value between 0 and 1. X_a denotes a $(k \times 1)$ vector of the pre-treatment features of the treated unit that we are trying to match in the best possible way and X_b denotes the $k \times J$ matrix of the same variables in the units of the donor pool. The purpose is to find the values of the X_b variables that are most similar to X_a , and the synthetic control, W^* , is the one minimizing the difference in the vector $X_a - X_b W$, which corresponds to the difference between the treated unit and a synthetic control. If $m = 1, \dots, k$, then X_{am} is the value of the variable m in the treated unit and X_{bm} is a $1 \times J$ vector with the values of the variable m in the donor pool. W^* is the value of W minimizing

$$\sum_{m=1}^k v_m (X_{am} - X_{bm} W)^2 \quad (1)$$

in which v_m is a weight for the relative contribution of each of the predictor variables.

Let Y_{jt} denote the outcome of unit j at time t ; then Y_1 denotes a $(T_1 \times 1)$ vector containing the values of the outcome for the treated period after the treatment, and Y_0 is a $(T_1 \times J)$ matrix containing the values of the outcome for the synthetic series. The impact of the treatment is obtained by comparing the post-treatment outcomes between the treated unit and the synthetic control:

$$Y_{1t} - \sum_{j=2}^{J+1} w_j^* Y_{jt} \quad (2)$$

A more detailed explanation of the method can be found in [Abadie and Gardeazabal \(2003\)](#) or [Abadie \(2021\)](#). The main purpose of any statistical exercise is to use inference to establish the validity of the results. This may seem difficult with the synthetic control method, which operates in a context in which some essential conditions for statistical inference are absent, such as the randomization of the sample and a choice of sample units through probabilistic means. However, the method allows for some exercises that provide equally valid results. These falsification exercises are termed “placebos” (drawing from medical terminology) and come in two types: in-time placebos and in-space placebos. Suppose we have found a strong impact of some intervention using synthetic control. We would question the result's robustness if a similar or greater impact were observed for a different time period. In order to control for this possibility, an in-time placebo test is implemented. This involves re-running the method, assigning the intervention to a different date. If no impact is detected, we can be confident that the date chosen for the intervention is appropriate. For example, in this study, we take the Carnation Revolution as having occurred in 1974. If we instead assign the intervention to 1964 and observe no significant effect, this supports the validity of 1974 as the actual relevant intervention year ([Abadie and Gardeazabal, 2003](#); [Abadie, 2021](#)).

We would also question the significance of the intervention if we were to find a similar or greater impact for the same date but on other members of the donor pool not exposed to the same kind of intervention. To assess this possibility, we reassign the intervention to the units in the donor pool that were not directly exposed to the intervention. If we find that the other units were not affected by it, we can be more certain that the intervention is unique to the treated unit. Statistical significance is then assessed by calculating p-values, derived by estimating placebo effects for each donor unit and computing the fraction of those that are equal to or exceed the effect observed for the treated unit. The resulting p-value represents the probability of observing a similar effect in another unit by chance ([Abadie and Gardeazabal, 2003](#); [Abadie, 2021](#)).

In our investigation, the Portuguese synthetic economy is obtained by applying the above procedure to a balanced panel sample of all countries that were members of OECD over the period 1955–2000: Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Iceland, Ireland, Italy, Japan, Luxembourg, Mexico, the Netherlands, New Zealand, Norway, South Korea, Sweden, Switzerland, Türkiye, the United Kingdom, and the United States, in addition to Portugal. This list of countries deserves some comments. The SCM requires the exclusion of units that experienced similar interventions to the treated unit. Two countries in our original sample posed potential problems in that regard: Greece and Spain, as they transitioned to democracy during these years, similarly to Portugal – Greece in 1974, and Spain in 1975 ([Prados de la Escosura et al., 1996, 2011](#); [Melcarne et al., 2021](#); [Alogoskoufis, 2021](#)), so they were removed from our sample. A note on the timeframe is also in order. Although our earlier discussion covered the broader postwar period, the empirical analysis is limited to 1955–2000 due to data availability. The year 1955 is the first in which we have comparable data for the large set of countries used in our study.

The outcome variables of interest are the level of GDP per capita (in real terms), the current account balance measured as a percentage of GDP, and the economy's investment-employment ratio. The treatment period begins in 1974. The variables used to compute the synthetic version of the Portuguese economy differed depending on the outcome variable taken into consideration. In the case of GDP per capita, the variables were consumption, investment, and Government consumption (all in per capita terms), under the logic that they are the components of GDP. Additional variables include welfare-relevant total factor productivity (TFP), trade openness, and the real effective exchange rate, all commonly recognized as determinants of long-term economic growth. In the case of the current account, the variables were the internal rate of return, the share of labor compensation in GDP, the real effective exchange rate, and the welfare-relevant TFP. These variables reflect the structural and external drivers of current account dynamics. The internal rate of return (IRR), for example, is a critical determinant of the current account because it reflects the profitability and attractiveness of investments within an economy. A higher IRR indicates that domestic projects yield better returns, which can attract foreign capital and reduce reliance on external borrowing. This inflow of foreign funds strengthens the current account by offsetting trade deficits or other imbalances. Moreover, the real exchange rate influences export competitiveness and emigrant remittances. In the case of the investment-employment ratio, relevant predictors include the internal rate of return, real exchange rate, welfare-relevant TFP, capital

depreciation rate (δ), and a human capital index. These variables were selected based on their theoretical relevance to investment dynamics. For instance, the complementarities between human capital and investments are acknowledged for a long time in the economics literature (Griliches, 1969; Goldin and Katz, 2008). In all cases, we maintain a consistent, parsimonious, and comparable set of predictor variables across the different outcome measures.

Data for GDP per capita, consumption per capita, investment per capita, government consumption per capita, investment-employment ratio, share of trade openness, welfare-relevant TFP, internal rate of return, capital depreciation, human capital and

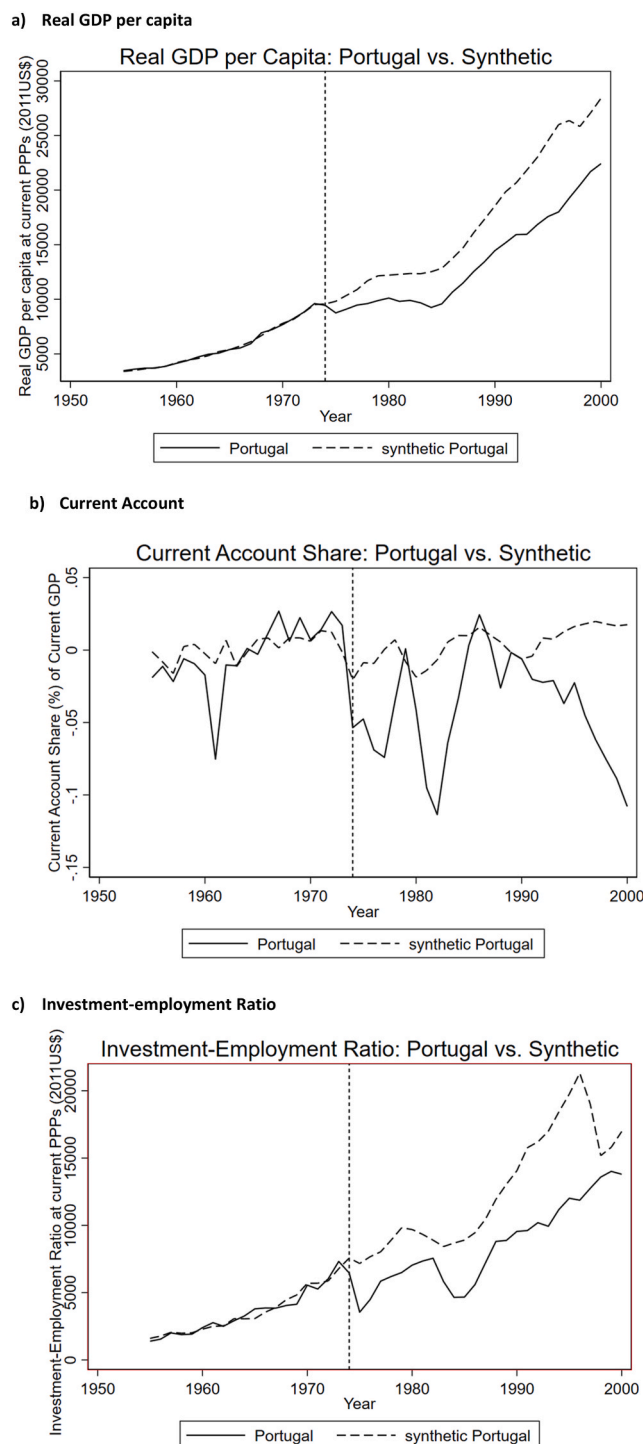


Fig. 3. GDP per capita, Current account, and capital-labor ratio: Portugal vs. synthetic (1955-1992), treatment: 1974.

share of labor compensation in GDP were sourced from PWT 9.1. Data for the current account were obtained in [Jordà et al. \(2019\)](#). Since this source did not have data for Portugal, we obtained it from [INE and BdP \(2021\)](#) instead. Data for the real exchange rate were drawn from the Bruegel Datasets. In the case of the current account, due to data limitations, the sample was restricted to a subset of 15 of the countries mentioned above, excluding Austria, Ireland, Israel, New Zealand, and South Korea. The description of these variables is available in [Table A.I in Appendix](#).

4. The impact of the Carnation Revolution on GDP per capita, the current account, and the investment-labor ratio

In this section, we present the results of the tests implemented to assess the impact of the Carnation Revolution and the instability that followed on the Portuguese economy. The section is divided in subsections, one for each variable: GDP per capita, current account, and investment-labor ratio. The reason for this division is to make clear the result of each test.

4.1. GDP per capita

[Fig. 3a](#) shows the performance of the Portuguese economy compared to its synthetic counterpart. Both declined in 1974, but while the synthetic economy began to recover in 1975, the actual Portuguese economy continued its downward trend, creating a structural break between the two. Although the Portuguese economy returned to growth after 1976, the pace was not sufficient to restore it to its previous trajectory, resulting in continued divergence from the synthetic series. This divergence became even more pronounced between 1980 and 1985. As time went by, we argue that the gap may reflect not only the initial revolutionary shock, but also the persistence of its institutional legacy — particularly the delayed reversal of nationalizations and the slower adoption of market-oriented reforms relative to other OECD countries. In this sense, delayed reform is not an independent confounding factor but part of the causal chain initiated in 1974. As discussed in [Section 2](#), it was only after 1985 that the Portuguese economy began to reform and recover more clearly, though it never caught up with the synthetic series by the end of the period in 2000. According to our estimates, Portuguese GDP per capita was almost 15% lower in 1976 than it would have been in the absence of the Carnation Revolution and the period of instability subsequent to it. This figure reflects the difference between the actual and synthetic GDP per capita levels in that year and can be considered the short-term impact of the revolution. By 2000, at the end of our period of analysis, the gap had widened to more than 40%, which we interpret as the long-term impact of the revolution at that time. We acknowledge, however, that the numbers for the long-term impact must be interpreted with caution. As we move away from the treatment date, the synthetic control may become less reliable, since post-treatment extrapolation increasingly depends on model assumptions rather than on the pre-treatment fit in which the donor pool was calibrated. The validity of the estimates thus potentially decreases over longer post-treatment windows, given that unit-specific trends, donor-pool drift, and time-varying confounders accumulate as we move further away from the shock.

Table 1

Country Weights for Synthetic Portugal, real GDP per capita, Current Account and Investment-employment ratio.

Country	(1) Synthetic Control Weight real GDP per capita	(2) Synthetic Control Weight Current account	(3) Synthetic Control Weight Investment employment ratio
Australia	0	0	0
Austria	0	na	0
Belgium	0	0	0
Canada	0	0.150	0
Denmark	0	0	0
Finland	0	0	0
France	0	0	0
Germany	0	0	0
Iceland	0	na	0
Ireland	0.163	na	0
Italy	0	0.254	0.186
Japan	0.312	0.435	0.197
Luxembourg	0.030	na	0
Mexico	0.070	na	0.107
Netherlands	0	0	0
New Zealand	0	na	0
Norway	0	0	0
South Korea	0.425	na	0.510
Sweden	0	0.161	0
Switzerland	0	0	0
Türkiye	0	na	0
United Kingdom	0	0	0
United States	0	0	0

Note: na means not available.

We can see in [Table I](#), column 1, the weight of the “donor” countries in the construction of the Portuguese synthetic counterfactual. The largest donor is South Korea, with a weight of 42.5%, followed by Japan, with a weight of 31.2%, Ireland, with a weight of 16.3%, Mexico, with a weight of 7%, and Luxembourg, with a weight of 3%. In one fundamental respect, the contributors to the donor pool make intuitive sense: much like Portugal, the most important (South Korea and Japan, which together account for almost 75% of the donor weight) were fast growing economies in the period from the 1950s to the mid-1970s. Portugal's growth path in the period has been compared in some literature to that of the so-called “Asian Tigers” (South Korea, Taiwan, and Hong Kong) ([Baklanof, 1992](#); [Mateus, 2006](#); [Amaral, 2005](#)), of which Japan is somehow a precursor. The inclusion of Ireland also makes sense, as it was a relatively less developed economy in the 1950s–1970s, undergoing a process of economic catch-up, albeit at a slower pace than Portugal and the Asian economies. [Table II](#), panel a shows that the synthetic control closely matches Portugal in terms of the predictor variables.

Falsification through placebo tests provides meaningful validation. We first apply the in-space placebo test, with results shown in [Table III](#), column 1. The overall probability of the effect being replicated with a different treated unit is 0%, according to the p-values displayed in the table. This can be interpreted, similarly to traditional statistical inference, as a 0% significance level, revealing the high adequacy of the exercise. This is confirmed by [Fig. 4a](#) and b, which illustrate the distinct post-treatment trajectory of the Portuguese economy and its clear divergence from the pre-treatment trend. We conclude that the observed effect could not have arisen from assigning the intervention to a different treated unit.

Next, we conduct the in-time placebo test, which is done by artificially reassigning the treatment to a date before the intervention. We rerun the model assuming the intervention took place in 1964, the midpoint of the pre-treatment period. [Fig. 4c](#) exhibits the results: synthetic Portugal closely reproduces the GDP per capita performance for the new pre-treatment period. Crucially, there is no new significant divergence in the trajectory between Portugal and its counterfactual before the Carnation Revolution, thereby suggesting that the estimated gap after 1974 is indeed determined by the treatment.

Finally, we provide results two further exercises. First, we include Greece and Spain, two economies that also transitioned to democracy during the mid-1970s and which, for that reason, were originally removed from our sample. As can be seen in [Fig. 4d](#), the results are similar to baseline. Second, we remove the richest per capita economy in our sample, Japan. The results depicted in [Fig. 4e](#) display a noisier pre-treatment fit, but seem to qualitatively confirm our baseline findings.

4.2. Current account

Turning now to the impact of the revolution on the balance of payments, our findings are reported in [Fig. 3b](#). Portugal experienced a serious international payments crisis beginning in 1974, while its synthetic counterpart suffered a much milder one. Also, synthetic Portugal returned to balance in 1975, whereas the actual Portuguese economy saw continued deterioration, starting to recover only in 1978 and completing its recovery in 1979. However, this was followed by a new and steeper deterioration until 1983. A renewed recovery began in 1984 and was achieved by 1985. From 1986 until 1992, the Portuguese economy maintained balance, replicating

Table 2
Predictor means for the pre-treatment period (1955–1973).

a) GDP per capita		
	Portugal	Synthetic Portugal
Consumption Per Capita	4017.888	3368.545
Investment Per Capita	1265.528	1515.985
Government Consumption Per Capita	618.7027	796.847
Trade Openness (share)	0.252	0.252
Real Effective Exchange Rate	0.951	0.968
Welfare-Relevant TFP	0.850	0.734
b) Current Account		
	Portugal	Synthetic Portugal
Internal Rate of Return	0.135	0.104
Share of Labor Compensation in GDP	0.643	0.642
Real Effective Exchange Rate	0.951	0.951
Welfare-Relevant TFP	0.850	0.782
c) Investment-employment Ratio		
	Portugal	Synthetic Portugal
Internal rate of return	0.135	0.127
Real Effective Exchange Rate	0.951	0.973
Welfare-Relevant TFP	0.850	0.773
Delta	0.026	0.031
Human Capital	1.343	1.998

Note: the mean for the OECD pool comprises all countries in the donor pool, for each outcome variable, as presented in [Table 1](#).

Table 3
P-values for in-space placebo effects.

Year	(1) GDP per capita	(2) Current Account	(3) Investment-employment ratio
1975	0	0.533	0
1976	0	0.133	0
1977	0	0.067	0.043
1978	0	0	0
1979	0	1	0
1980	0	0.533	0
1981	0	0	0.043
1982	0	0	0.174
1983	0	0	0
1984	0	0.333	0
1985	0	0.8	0
1986	0	0.6	0
1987	0	0.8	0.043
1988	0	0.467	0.087
1989	0	0.8	0.043
1990	0	0.933	0
1991	0	1	0
1992	0	0.4	0
1993	0	0.667	0
1994	0	0.333	0
1995	0	0.667	0
1996	0	0.333	0
1997	0	0.2	0
1998	0.417	0.067	0.913
1999	0.417	0.267	0.913
2000	0.417	0.267	0.261

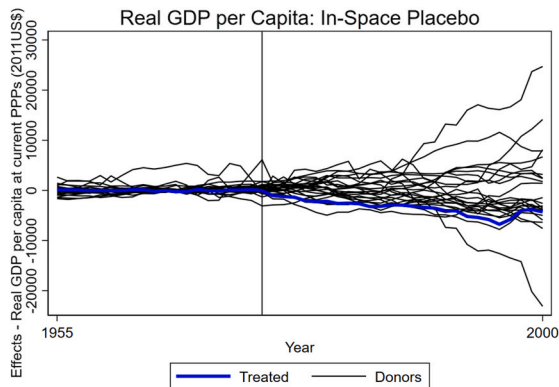
the synthetic, but by this moment a previous decade of persistent imbalance had already been felt. And, showing the fragility of such balance and the persistent post-revolutionary difficulty in balancing external accounts, a new decade of deterioration begun in 1992. The relevant donor countries in this case are Japan, with a weight of 43.5%, Italy, with a weight of 25.4%, Sweden, with a weight of 16.1%, and Canada, with a weight of 15% (Table I, column 2). As noted, when commenting on the results for GDP per capita, the large contribution of Japan confirms the Asian Tiger profile of the Portuguese economy before 1974. Indeed, Japan, the Asian Tigers, and Portugal enjoyed in those decades a growth process closely linked with increased openness and a healthy balance of payments. Contrary to Japan and the Asian Tigers, Portugal's consistently positive external balance did not result from an outstanding export success (although there was some), but mostly from the massive influx of emigrant remittances. However, in terms of the shape of the current account, one process is indistinguishable from the other, meaning that Japan should still be viewed as being a reasonable comparator. Table II, panel b shows that the synthetic controls are reasonably close to Portugal in terms of the predictor variables. Nevertheless, we acknowledge that the pre-treatment fit is not as tightly estimated as for the GDP per capita and therefore the above mentioned results must be interpreted with a grain of salt. The table also provides a comparison with the possible alternative method of using the same country sample with their contributions arising from an unweighted average. The synthetic exercise gives similar results.

When we break down the results by individual years following the intervention, the years of 1976, 1977, 1978, 1981, 1982, and 1983 show the strongest significance. These were precisely the years in which the Portuguese current account diverged most markedly from that of the synthetic control. Fig. 5a and b illustrate the uniqueness of the Portuguese path in relation to the donor units. Fig. 5c has the results of the in-time placebo when the intervention is reassigned to 1964: synthetic Portugal reproduces very closely the current account behavior for the new pre-treatment period. Importantly, no significant divergence emerges, reinforcing the conclusion that the post-1974 gap is indeed causally linked to the revolution.

4.3. Investment-employment ratio

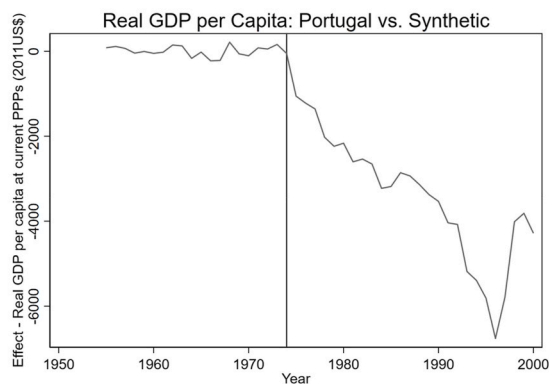
As for the impact of the revolution and the subsequent instability on investment per employed person, we can see in Fig. 3c that, *vis-à-vis* the synthetic counterfactual, the effect was persistently negative. The revolution caused a structural break in capital accumulation in the Portuguese economy, leading to a lower growth path than in the preceding 20 years, and lower than it would have been in the absence of the revolution. This is a decisive contribution to explain the consistently lower growth performance of the Portuguese economy in the long-term. The relevant donor countries for this exercise are South Korea, with a 51% contribution, followed by Japan, with a weight of 19.7%, Italy, with 18.6%, and Mexico, with 10.7%. Once again, we find a large contribution coming from South Korea and Japan, confirming the “Asian Tiger” profile of the Portuguese growth process before 1974. The inclusion of Italy further suggests certain underlying similarities between its economy and that of Portugal. Table II, panel c shows that the synthetic control series align closely with Portugal in terms of the predictor variables, with the slight exception of human capital, with donor economies being, on

a) Portugal vs. donors

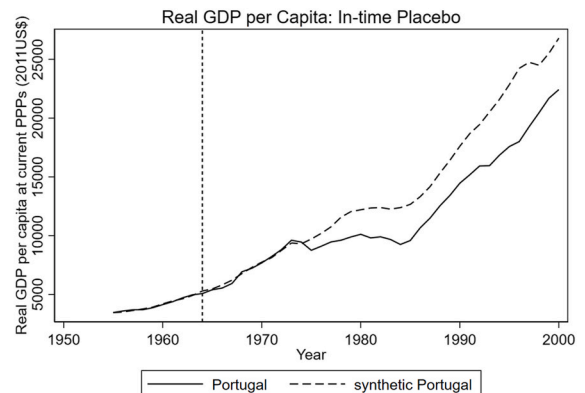


Note: Calculations computed using Galiani and Quistorff (2017).

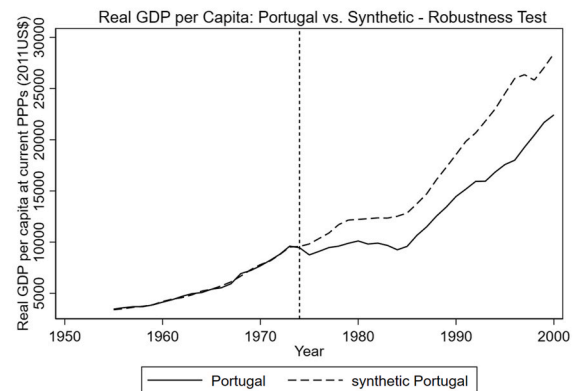
b) Gap post-1974 Portugal vs synthetic



c) Treatment relocation 1964

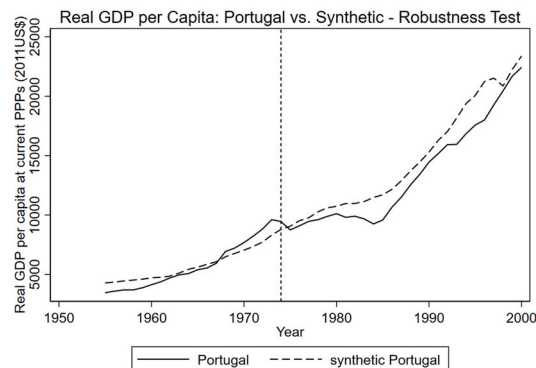


d) Adding Greece and Spain



Note: Donors weights are the same as in the baseline results.

e) Removing Japan from baseline results



Note: Donors weights are South Korea (54%), Mexico (23%), the Netherlands (16%), and Australia (6%).

Fig. 4. Placebo tests synthetic control method real GDP per capita Portugal (1955–1992), treatment: 1974

Note: Calculations computed using Galiani and Quistorff (2017).

Note: Donors weights are the same as in the baseline results.

Note: Donors weights are South Korea (54%), Mexico (23%), the Netherlands (16%), and Australia (6%).

average, more educated than the Portuguese one.

In 1982, investment recovered somewhat, which is visible in Fig. 6c, with no significant differences to the counterfactual. But, after 1983, the Portuguese path becomes again considerably different. Then, in the final years of the series, the differences are not significant again, with high p-values in the late 1990s. Fig. 6a and b shows the Portuguese path in relation to the donor units, which is quite specific in the years 1975–1979, and 1983–1986, explaining the structural break visible in the test. Fig. 6c displays the results of

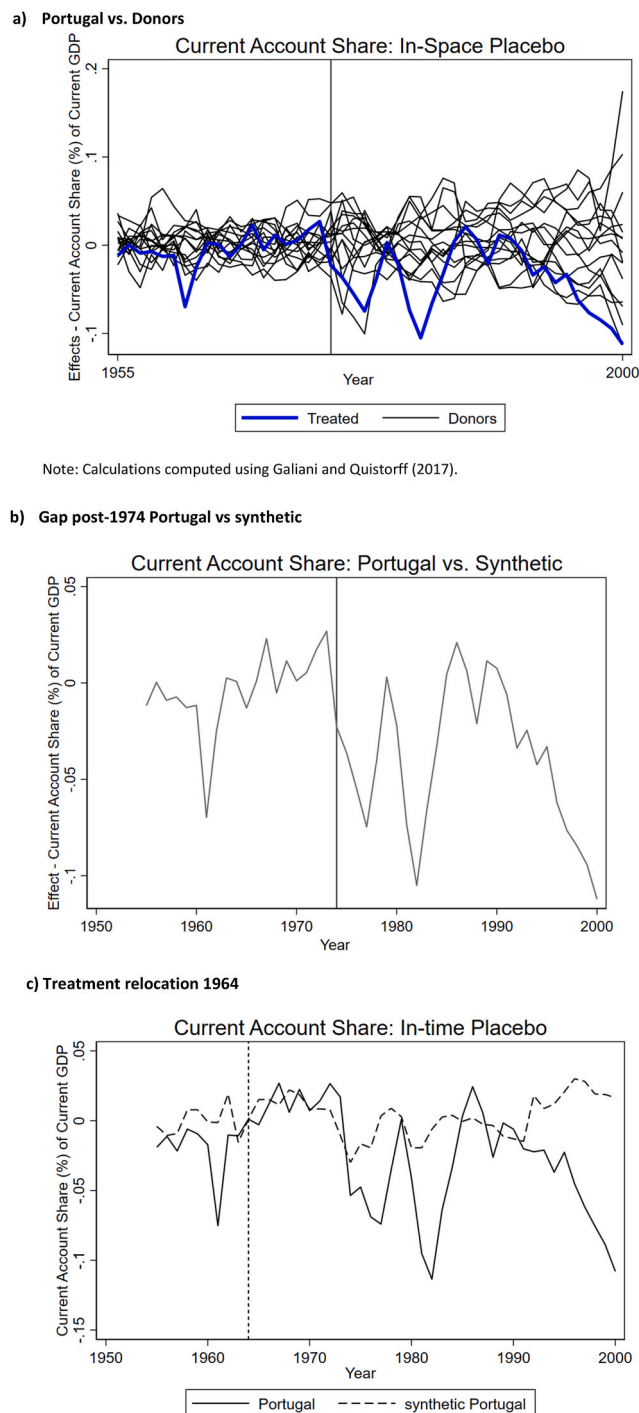
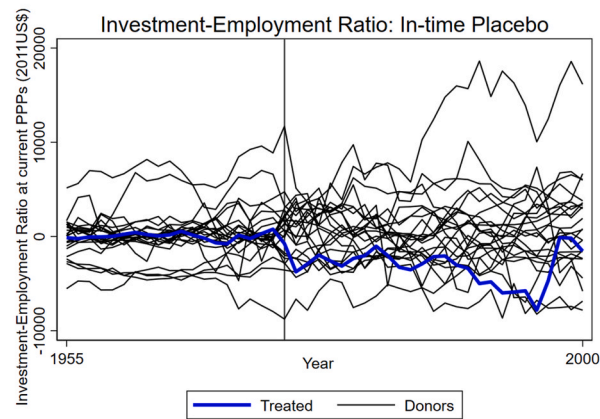


Fig. 5. Placebo tests synthetic control method current account Portugal (1955-1992), treatment: 1974.

the in-time placebo when the intervention is reassigned to 1964: synthetic Portugal reproduces the investment-employment ratio behavior for the new pre-treatment period. Most importantly the post-intervention divergence is minimal, reinforcing the conclusions that there were no anticipation effects and that the observed gap after 1974 is attributable to the treatment.

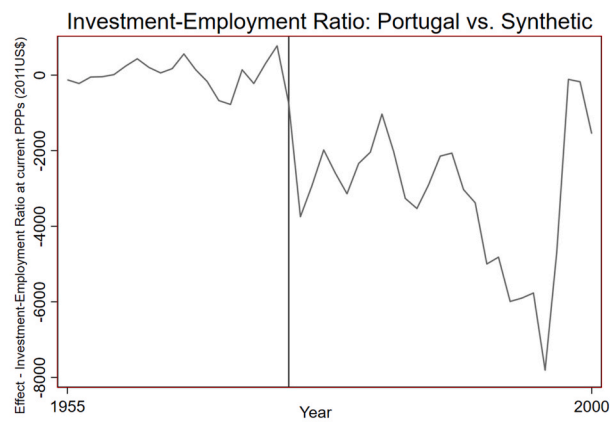
The three tests above allow support the conclusion that the political and economic events following the Carnation Revolution worsened Portugal's situation compared to a scenario in which the country had been affected only by the international crisis. Ultimately, the dilemma between rebalancing the economy externally and making it grow took a decade to be resolved. Moreover, the revolution brought negative structural change to the process of capital accumulation, leading to a lower growth path in the Portuguese

a) Portugal vs. Donors



Note: Calculations computed using Galiani and Quistorff (2017).

b) Gap post-1974 Portugal vs synthetic



c) Treatment relocation 1964

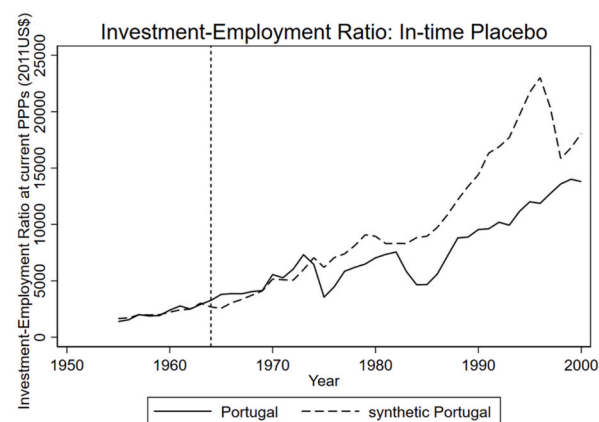


Fig. 6. Placebo tests synthetic control method Investment-employment ratio Portugal (1955-1992), treatment: 1974.

economy.

5. Conclusion

We used the synthetic control method to measure the causal economic impact of the Carnation Revolution in Portugal, to separate it

from the impact of the 1970s international crisis. The Revolution unleashed a series of political, social, and economic tensions that made the economic situation of the country extremely difficult. These included a wage explosion, a long-term increase in labor costs, capital flight, expropriation of the assets of the business elite, the arrival of about 500,000 to 700,000 refugees from the empire, less emigrants' remittances, and fewer tourist visits. According to our findings, this set of factors determined a steeper fall in GDP per capita after 1974 than what it would have happened had the country's economy been affected only by the 1970s international crisis. The effects were felt in the short-term, but also in the long-term, thanks to capital flight and a labor shock. Both affected the factor mix of the economy negatively, putting it on a lower growth path than what would have happened without the revolution and the instability that followed. The socioeconomic events associated with the revolution were also responsible for a more severe current account crisis than the one that would have resulted from the impact merely of the international crisis, something that made the recovery more difficult. A dilemma for the country's policymakers was created, as they had to decide between two opposing objectives: returning to growth or rebalancing the economy externally. This means that the economy's external constraint posed limits to its performance: attempts at kickstarting growth brought external imbalance; attempts at correcting such imbalance slowed down growth. This policy trade-off plagued the Portuguese economy for a full decade.

Our paper makes an empirical contribution to the theory of revolutionary outcomes and helps refine some of its propositions, particularly those concerning the question of why some revolutions result in a transition from authoritarianism to constitutional democracy, while others merely replace one form of authoritarianism with another. Where the existing literature often categorizes revolutions as initiated either by radicals or moderates, we argue that the Carnation Revolution was initiated by both, leading to a complex and drawn-out process of political clarification. Even if in the end moderation prevailed, a radical legacy was also embedded in the resulting political and economic system, with a serious impact on the democratic regime's future, with significant implications for the future of the democratic regime, particularly in terms of its economic prospects. Revolutionary outcomes, therefore, need not be strictly binary; they can combine elements of both radicalization and moderation. Our results should therefore not be interpreted as evidence that democratization per se reduces economic performance. Rather, they reflect the economic consequences of abrupt institutional disruption and radical policy experimentation, capital expropriation, and macroeconomic instability during the revolutionary period.

One singularity of this study is that 1973 Portugal was somewhat unique among small states in still having a substantial colonial empire. Another important remark is that we do not examine how income inequality, gender inequality, or access to public goods such as education, healthcare, and justice evolved following the Carnation Revolution. We leave these important topics as avenues for future research.

CRediT authorship contribution statement

Luciano Amaral: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Bruno Lopes Marques:** Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **João Pereira dos Santos:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

We report that we do not have any conflict of interests.

Appendix

Table A.1

Variable	Description	Source
Current Account	Share of GDP (Current Account/GDP)	Portugal: INE and Bdp (2021) Other countries: Macrofinancial History.
Real GDP per Capita	PPP, 2011US\$ (Expenditure-side real GDP at current PPPs (in mil. 2011US\$)/Population in millions)	Penn World Table
Investment-employment Ratio	PPP, 2011US\$ (Share of gross capital formation at current PPPs) * Output-side real GDP at current PPPs (in mil. 2011US\$)/(Number of persons engaged (in millions))	Penn World Table
Consumption Per Capita	PPP, 2011USD (Share of household consumption at current PPPs * Output-side real GDP at current PPPs (in mil. 2011US\$))/Population in millions)	Penn World Table
Investment Per Capita	PPP, 2011USD (Share of gross capital formation at current PPPs * Output-side real GDP at current PPPs (in mil. 2011US\$))/Population in millions)	Penn World Table
Government Consumption Per Capita	PPP, 2011USD (Share of government consumption at current PPPs * Output-side real GDP at current PPPs (in mil. 2011US\$))/Population in millions)	Penn World Table
Trade Openness	Share of Expenditure-side real GDP PPP, 2011USD (Share of merchandise exports at current PPPs - Share of merchandise imports at current PPPs (negative share in Penn World Table))	Penn World Table

(continued on next page)

Table A.I (continued)

Variable	Description	Source
Welfare-relevant TFP	PPP, 2011USD USA in 2011 = 1	Penn World Table
Real Effective Exchange Rate	Annual, Index Basket of 66 countries Rescaled to 1970	Bruegel Datasets
Internal Rate of Return	Real Internal Rate of Return on Capital (GDP - Labor Income - Natural Resource Rents) Jorgenson and Nishimizu (1978)	Penn World Table
Delta	Average depreciation rate of the capital stock	Penn World Table
Human Capital	Human capital index, based on years of schooling and returns to education	Penn World Table
Share of Labor Compensation in GDP	Current National Prices	Penn World Table

Data availability

Data will be made available on request.

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